MATCHED 10-MILLION-TOKEN STUDY

Official nanoGPT vs. Proven Pure Hierarchy

A controlled comparison of predictive quality, long-context behavior, and per-sequence inference state under the same data, teacher objective, training exposure, and evaluation protocol.

11.64%
LOWER HIERARCHY CE AT 2,048

14,002.6x
LARGER NANOGPT KV STATE AT 2,048

99.9929%
FEWER HIERARCHY STATE BYTES

At context 2,048, the hierarchy achieved CE 5.291337 versus nanoGPT 5.988185 while using 5.375 KiB rather than 73.5 MiB of analytic BF16 sequence state.

Boundary of the result

This is not a universal quality win. nanoGPT is better at the trained 512-token context and has a slightly higher four-task benchmark mean. The hierarchy advantage appears at the 1,024- and 2,048-token evaluations, where its fixed recurrent state is also thousands of times smaller than nanoGPT's theoretical KV state.

Report date 2026-08-15 | Status: training, fixed evaluation, and benchmark complete

A matched experimental contract

Same exposure. Same data. Near-identical scale.

The comparison keeps the proven hierarchy design unchanged and evaluates a pinned official nanoGPT comparator under the historical Aug-8 teacher-distillation protocol.

Variable	Matched value	Why it matters
Teacher	Qwen3-0.6B-Base	Same target distributions
Training / validation data	Byte-identical train.bin / val.bin	Same corpus bytes
Training context	512 tokens	Same token window
Batch / world size	4 sequences / 1 process	Same examples per step
Optimization	4,883 steps / 10,000,384 tokens	Same exposure
AdamW	LR 0.0003; warmup 500; betas 0.9/0.95	Same schedule
Objective	0.1 next-token CE + 1.0 forward KL	Same loss contract
Training / eval seeds	12001 / 14001	Same deterministic schedule
Fixed evaluation	512, 1,024, and 2,048 contexts	Same held-out windows

Parameter match

Model	Stored parameters	Architecture
Pure hierarchy	119,694,800	Original width-688 recurrent hierarchy
Official nanoGPT	119,685,440	Width 448 / 21 layers / 7 heads
Absolute difference	9,360 (0.00782%)	Effectively parameter matched

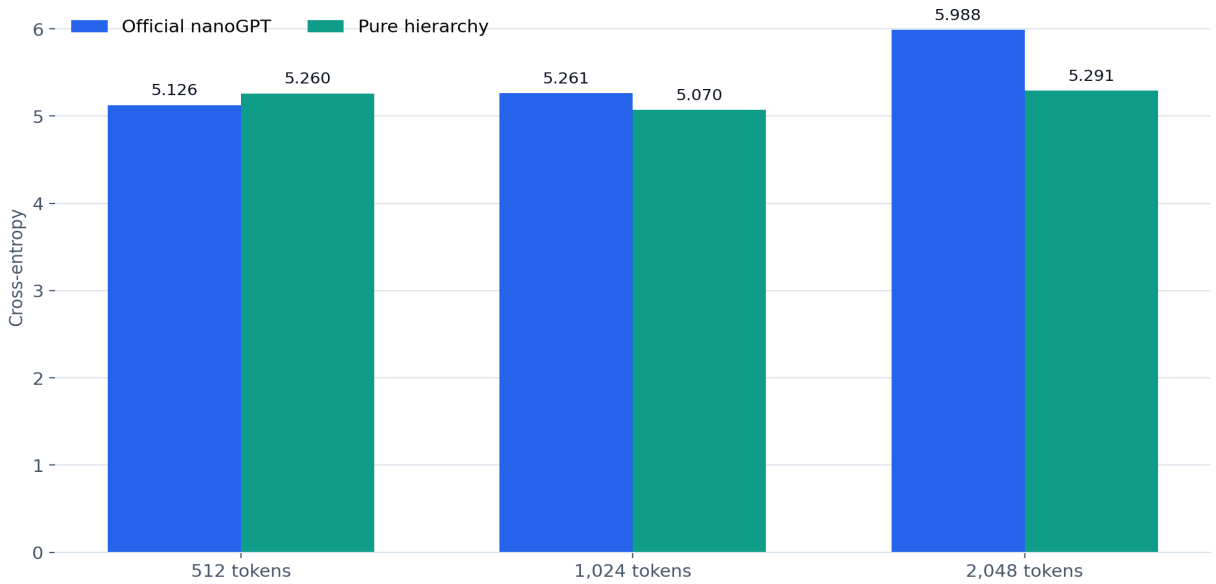
Design boundary

The comparator uses upstream nanoGPT model bytes pinned to commit 3adf61e154c3fe3fca428ad6bc3818b27a3b8291. No perturbation solver, post-training optimizer, extra objective, or alternate hierarchy design was introduced.

PREDICTIVE QUALITY

nanoGPT leads at 512; hierarchy leads at longer contexts

Fixed-evaluation cross-entropy. Lower is better; bars use a zero baseline.



Context	nanoGPT CE	Hierarchy CE	Outcome	Evaluation shape
512	5.126457	5.260253	nanoGPT 2.54% lower	16 batches x 2 sequences
1,024	5.261362	5.070329	Hierarchy 3.63% lower	2 batches x 1 sequence
2,048	5.988185	5.291337	Hierarchy 11.64% lower	2 batches x 1 sequence

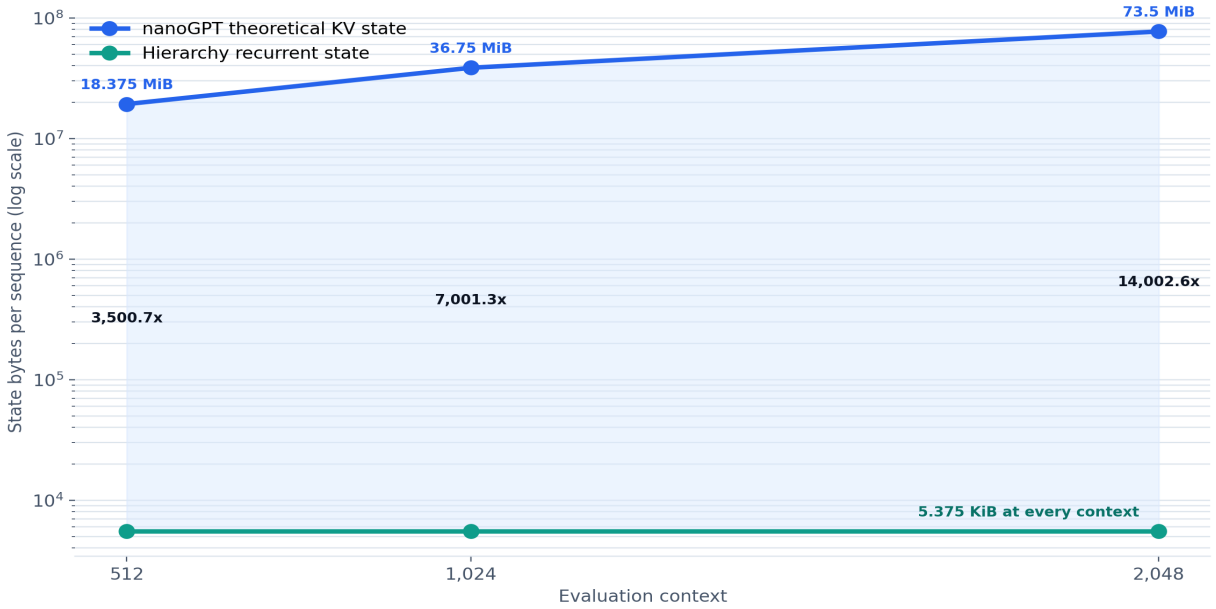
Interpretation

nanoGPT's fixed CE is 2.54% lower at the trained 512-token context. The hierarchy reverses the result at 1,024 tokens and widens the advantage at 2,048 tokens. The long-context samples are small, so these points should be treated as strong directional evidence rather than a confidence-bounded population estimate.

STATE SCALING

Fixed recurrent state versus linearly growing KV state

Per-sequence BF16 state on a log scale. This is architecture accounting, not total GPU memory.



Context	nanoGPT KV state	Hierarchy state	nanoGPT / hierarchy	Hierarchy reduction
512	18.375 MiB	5.375 KiB	3,500.65x	99.9714% fewer
1,024	36.75 MiB	5.375 KiB	7,001.30x	99.9857% fewer
2,048	73.5 MiB	5.375 KiB	14,002.60x	99.9929% fewer

Architecture	Analytic BF16 state formula
Pure hierarchy	4 recurrent tensors x width 688 x 2 bytes = 5,504 bytes
Official nanoGPT	K/V x 21 layers x 7 heads x 64 head dimension x context x 2 bytes

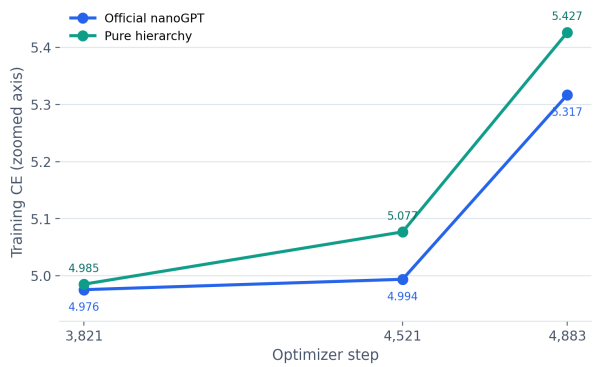
What this does and does not mean

These are nominal per-sequence state values. They do not include parameters, activations, optimizer state, allocator overhead, or total runtime memory. The nanoGPT adapter did not directly materialize or measure an incremental KV cache; its values are architecture accounting.

SUPPORTING RESULTS

Matched checkpoints and downstream benchmark

Matched training-minibatch CE



Four-task benchmark (limit 1,000)



Matched deterministic training minibatches

Step	Tokens	nanoGPT CE	Hierarchy CE	Lower CE
3,821	7,825,408	4.975600	4.985255	nanoGPT 0.19%
4,521	9,259,008	4.993795	5.076864	nanoGPT 1.64%
4,883	10,000,384	5.317025	5.426507	nanoGPT 2.02%

nanoGPT rows come from the included 246-record training log. Historical rows are preserved transcript values; the primary conclusion uses fixed evaluation.

Four-task benchmark

Task	nanoGPT	Hierarchy	Higher score
ARC-Easy	0.340351	0.329825	nanoGPT
HellaSwag	0.275000	0.257000	nanoGPT
LAMBADA	0.000000	0.000000	Tie
PIQA	0.530000	0.534000	Hierarchy
Mean	0.286338	0.280206	nanoGPT

nanoGPT's benchmark mean is 0.006132 points higher, a 2.19% relative advantage. The benchmark is one seed with no confidence interval.

INTERPRETATION

The result is a long-context state-efficiency result

1	**At 512 tokens** nanoGPT has 2.54% lower fixed CE and is therefore the stronger model at the trained context.
2	**At 1,024 tokens** Hierarchy has 3.63% lower CE while nanoGPT's theoretical state is 7,001.3x as large.
3	**At 2,048 tokens** Hierarchy has 11.64% lower CE while nanoGPT's theoretical state is 14,002.6x as large.
4	**Across the benchmark** nanoGPT has a modest 2.19% relative mean advantage; PIQA favors hierarchy, ARC-Easy and HellaSwag favor nanoGPT, and LAMBADA ties at zero.

Bottom line

At context 2,048, the proven hierarchy combines lower CE with 99.9929% fewer analytic state bytes. nanoGPT remains the better result at the trained 512-token context and on the downstream benchmark mean. Both facts are necessary for an accurate conclusion.

PROVENANCE

Content-addressed result evidence

The hashes below bind the report to the historical hierarchy artifacts, the frozen official nanoGPT package, the completed training output, and the downstream benchmark output.

Artifact	SHA-256 / commit
Hierarchy config	`f47d8cabafa5f16ff32c1169fba2064bf8124907cfc99e7d8374212cc25c58fa`
Hierarchy result	`ffe3dea20685e88381eab65f02bba504e6870975a0ec06854de73718b88f769d`
Hierarchy benchmark	`109f61c95d03dba065a31c4cbc683e7b96ebb52a507afab9cae8136e94188b43`
Hierarchy checkpoint	`beb12f7a9fce10195c639365cc673200e04c97affa9557239b70edd28bd9c66f`
nanoGPT frozen manifest	`dc4e324d7f43ddf417c6641ce3c6ec856cbec69a6d1b83e93f2fe5b0690c833c`
nanoGPT result	`2120eac7d5f699d810425a8b793eccc0fd80674430751a2cc6d33361d81e3d44`
nanoGPT training completion	`6e3c1eda103ff89e6ae687e800f2cbf3c0cb76dd3a25b12fb2bc68c0df9419a7`
nanoGPT final model	`7de4e9d81b606ec6b0d7e9fbaf4dcdcb30aa55456e775a07371aaf65a6591241`
nanoGPT training log	`867e64674ed291d3eb8e8594d316c68c19c830fc7a70206d9542680ddcce532c`
nanoGPT benchmark	`1df9a3e0f410ffee5189e111137acd2755717a6a7c81cfb099e54c4cbb7f340e`
nanoGPT benchmark completion	`fc28be724d939e0f619d935d47b1ffc356a06e2c66b6530efd41523ca3042be6`
Upstream nanoGPT commit	`3adf61e154c3fe3fca428ad6bc3818b27a3b8291`
Pinned nanoGPT model.py	`7c01703240dbec5d554527dc666e35b3df8391d0b117fddc07afcf325a21d11c`
Training data	`834c82d2ea76e2f17aa43d15862f5ef444322c37d9b29d07fe803bb21ac163e4`
Validation data	`8517985c4979dbc95c83137ee534e30e789e2275e4ba1ee1fd31e1f0d566e4d3`
Teacher weights	`cd2a512003e2f9f3cd3c32a9c3573f820bb28c940f73c57b1ddaa983d9223eba`

Source report

report/source_report.md

`Source report SHA-256: b26912fa75a3bccca3d1ac3b7110708b366c0b8c565d99c3161e1b5e4a88b3b9`

Prepared from immutable experiment outputs. All numerical state comparisons are analytic BF16 per-sequence architecture accounting.